“Does Medium-term Price Momentum predict future Equity Returns?”

1. Introduction

Momentum is the concept of stocks rising or falling rapidly, usually due to investor expectations or high trading volumes – in this project I will aim to settle the hypothesis of whether medium-term momentum (12 months) does play a factor towards stock returns, and will prove this by statistical and performance testing. Jegadeesh & Titman in 1993 proved this trading strategy through testing it on the NYSE and AMEX from 1965-1989. Through testing various momentum-based strategies they saw significant returns, ranging from 1.19 to 2.43. They also found that holding momentum-driven stocks for shorter holding periods seemed to produce higher returns – my strategy for this project will aim to prove this, and will involve weighting top-performing stocks each month over the holding period, such that my portfolio at any given time will consist of the holding period multiplied by the number of the top 10% performing stocks each month. Likewise with Jegadeesh and Titman, I will define medium-term momentum as the past 12 months performance of stocks, bar the previous month, which will aim to capture investors’ underreaction and avoids short-term and long-term noise.

The project will be split out in 5 sections: Initially, I will collect the data needed to test this hypothesis, before then cleaning and standardising data. After this, I will code the signal of my definition of momentum, and apply it to my cleaned dataset. Using this new data column of momentum, I will then analyse the top performing stocks of each month to build my portfolio, which I will use to collect summary statistics and performance data outlining whether or not my hypothesis is statistically significant or not.

2. Data Collection and Cleaning

I chose to test this strategy from the S&P 500 universe of stocks, ranging from 2010 to present (Oct 2026) – one reason is due to the data quality and availability. Price data from large cap companies are clean and easy to pull from data APIs. Moreover, the S&P 500 is regarded as a standard benchmark, where many other strategies have been tested against it, so the effectiveness of my method will be a natural comparison for performance. However, a limitation of choosing the current S&P 500 from 2010-2026 is the survivorship bias - these stocks have all ‘survived’ over the past decade or so, therefore they are likely to perform well in the long-term compared to stocks that may have collapsed that were part of the S&P 500 10 years ago. Pulling data from the yfinance API, I was able to import all tickers from the S&P into a large dataset. Every stock data came with the columns raw close, adjusted close or raw volumes as some examples – looking into it, I chose to use the adjusted close column as the basis for my stock analysis. Adjusted close accounts for adjustments made to the stock price which wouldn’t reflect an actual change in return. For example, stock splits e.g. 2 for 1 would see the share price down by 50%, when this would be a false crash. Therefore, I collected all adjusted price data from

the S&P 500 – to clean this dataset, I looked at the number of zero and duplicated columns for each stock. As negative price values are also invalid, I tested against this. Furthermore, I checked for suspicious returns. For example, when returns were 'suspicious', which I deemed more than 50% in one day (since all companies are large cap), or invalid, where returns were greater than -100%, then I also checked for these returns.

After cleaning the data, I pulled the monthly data from the large, raw dataset and gave it a new column 'dollar volume', which is the monthly price multiplied by the monthly volume bought. I then added a penny stock filter, where I set raw_prices_m >= 5, which keeps stock months where the unadjusted price is at least $5, as low stock months have wide spreads and noisy returns. Furthermore, I added a liquidity filter, keeping stock months where more than $1m was traded over the month, which screens out stocks you couldn't realistically buy or sell without moving the price.

3. Statistical Testing of Portfolio Returns

3.1 Momentum signal and portfolio construction

To build the momentum signal, I took the last 2-12 monthly returns for each stock and multiplied them all together to give an overall momentum signal for a stock for each month, excluding its first 1-11 months. From this, I then built my strategy. Taking each month's top and bottom 10% performing stocks of medium momentum, my strategy aimed to buy the top performers and short the bottom ones. I assigned each to a respective list. Then, for all stocks, I computed its returns for the next k months, where k is the holding period, labelling this as growth – after this, I averaged all these returns and subtracted the bottom from the top to get the long-short return.

Using this function, I then implemented the Jegadeesh-Titman overlapping portfolio approach – with the holding period of k months, each month's portfolio return would be weighted up to k months for each month. For example, having a holding period of 3 months would mean you would have held 3 portfolios at a time, one for each of the past three months, where returns would have been accumulated. These portfolios are then averaged to achieve the final portfolio return. To optimise the holding period, I collected the portfolio return for 1,3,6 and 12 month holding periods.

3.2 Testing Approach

I then set out to test the effectiveness of my strategy using statistical testing. I firstly used a t-test, which assumed an initial hypothesis that the sample mean = 0, however I used the HAC form as my portfolio return series showed heteroskedasticity, due to returns being more volatile than others, and autocorrelation, as portfolios would overlap over each other every month. I also used a Wilcoxon signed-rank test, which tests the initial hypothesis that 0 is the median of the portfolio returns. Therefore, if either were significant, it would show that the method is indeed effective to some extent. I then tested its overall performance as shown on the graph.

However, a limitation of using the Wilcoxon signed-rank test is that it assumes independent observations – this is significant as my portfolios are built from overlapping cohorts, so monthly returns are positively autocorrelated. This is corrected by the Newey-West t-test, which is why I used it.

From my portfolio strategy, even across 1,3, 6 and 12 holding months, there was no significant evidence that medium momentum had an effect on future returns. For example, with a holding period of 1 month, the mean was 0.0037 (+0.37%) with a t-statistic of 0.85, which shows insignificance at the 5% level. The Newey-West p-value, corrected for autocorrelation, was also 0.40, clearly demonstrating that the results were fairly likely given a null hypothesis of a 0 mean. The Wilcoxon p-value was also 0.3447, which also fails to reject the null hypothesis that the returns are centred on 0. The Sharpe ratio was 0.21, which measures return per unit of risk, is poor – the equity market has historically delivered 0.4-0.5 in the long-run, so the sample can't distinguish between "no momentum" and "a modest momentum premium".

For holding months higher than 1, although the minimum portfolio loss was smaller in magnitude, most other statistics fell – the mean monthly return, the p-value for the Newey-West t-test increased alongside the Wilcoxon to 0.9845 and 0.4690 respectively for the 12-month holding period. The Sharpe ratio was also low at 0.21 for a 1-month holding period - this also fell as the holding period increased, highlighting that medium-momentum is a stronger indicator for shorter-term returns.

4. Performance and Risk Analysis

The Sortino ratio, also low, also fell as k increased, going from 0.305 to 0.006 for the 1 to 12 month holding periods, confirming that lower holding months produce better performance relative to their downside risk. However, with lower holding months, there is higher volatility where it is 0.21 for a 1-month holding period to 0.004 for a 12-month holding period – this is likely due to the overlapping portfolios from each month so there is less risk through the buying of a more diversified portfolio, although mean return still fell higher than the volatility causing the Sharpe ratio to also decline as the holding period increased.

The maximum drawdown (largest fall from one month return to the next) was -0.496 for a 1-month holding period. Compared to a market-neutral strategy earning a geometric return of about 2% at best, that is a very poor trade-off, summarised by the Calmar ratio at 0.045 (showing the return earned per unit of worst-case loss).

The plot in my performance testing section shows the cumulative growth of 1 unit invested in my strategy for different k-holding months. One feature that is immediately noticeable is that the lines move together – the 1,3 and 6 lines seem to follow each other very closely, showing that overlapping cohorts hold largely the same stocks. The maximum drawdown I described earlier is also visible, where around month 125 to 165 there is a large fall of roughly 50%, which is consistent with

my results. Overall, however, little net progress is made, where a neutral arithmetic return of 2% would still beat this strategy.

One limitation of my results was that no transaction costs, shorting costs or financing were taken into account – net performance would even be lower than my gross figures, as it costs capital to trade.

5. Conclusion

In conclusion, I failed to achieve any meaningful results, with both my null hypotheses of the median and mean of my portfolio returns being 0 were not able to be disproven. My test statistics were insignificant, however I did find somewhat interesting insights. There was noticeable evidence that medium-momentum was a stronger indicator to returns in the short-term i.e. 1 month compared to 6 or 12 months which is also consistent with Jegadeesh and Titman. Although the volatility decreased, the mean return fell even quicker as the holding period increased, causing the Sharpe and Sortino ratios to also fall, demonstrating the increased risk accompanied with holding the portfolio for a longer period.

However, the limitations of my method may offset the accuracy of my results – for example, using a Wilcoxon signed rank test did not take into account autocorrelation, which was a significant aspect of building the portfolio based on the Jegadeep-Titman strategy. Moreover, Jegadeep and Titman managed to prove a significance of around 1.5% from this strategy through using the NYSE and AMEX from 1965 to 1989 – likewise, I checked robustness through using e.g. penny-stock and other liquidity filters, but their universe was much larger and based on CRSP, which included delisted stocks so there was not a survivorship bias like mine. My time period also spanned around 15 years compared to their 25 years.

All in all, although I did not find significant evidence to disprove my null hypotheses, it was interesting to see how my results compared to Jegadeesh and Titman's, and I am eager to continue exploring my passion for the stock market.